

ORIENTATION DISCRIMINATION TASK

PSY310: Lab in Psychology

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GitHub Link: <https://github.com/evaj-beep/PSY310-LAB-IN-PSYCH.git>

Introduction

An Orientation Discrimination Task is a sensory test that is used to measure how accurately an individual (or an animal) detects the smallest of changes in an angle or the direction of two visual lines or physical surfaces (Vázquez et al., 2000).

The participants in this experiment are made to view simple visual patterns such as lines or Gabor patches which are striped, fuzzy, circular patches of light. The participant is shown the stimulus side by side and then once the test starts, must state whether the stimulus is tilted clockwise or counterclockwise with reference to the first stimulus shown.

This task measures the participants sensory acuity and cognitive processing. This in turn evaluates how well the primary visual cortex processes spatial details by finding the 'just noticeable difference' between the lines or Gabor patches (Mikellidou et al., 2023).

Method

Participants

The study involved a convenience sample of one female university student. The participant was 20 years old. They were recruited through personal peer contacts at Ahmedabad University. Prior to participation, the subject provided informed consent, and all acquired data was anonymised to safeguard confidentiality.

Materials

- **Hardware:** Stimuli were presented on a Lenovo IdeaPad Gaming laptop equipped with an integrated display monitor and keyboard. Participant responses were recorded via the built-in keyboard, using the 'up' key and the 'down' key to indicate change in orientation or tilt of the grating.
- **Software:** The task was programmed, presented, and recorded using PsychoPy (2026 version). PsychoPy controlled the presentation timing of each stimulus, randomized trial presentation, and automatically logged both response accuracy and response latency (reaction time) for each trial.
- **Stimuli:** Gaussian-windowed sinusoidal grating patches, also known as Gabor patches, were displayed prominently on the display screen as visual stimuli. The grating patch's orientation was changed along a baseline on each trial, resulting in either vertical or slanted orientations. Participants were told to press the "up" arrow key when the grating was slanted clockwise ("up") and the "down" arrow key when it was tilted counterclockwise ("down"). For later signal detection analysis, the experimental program automatically recorded the participant's precise key press as well as the accuracy of their answer.

Procedure

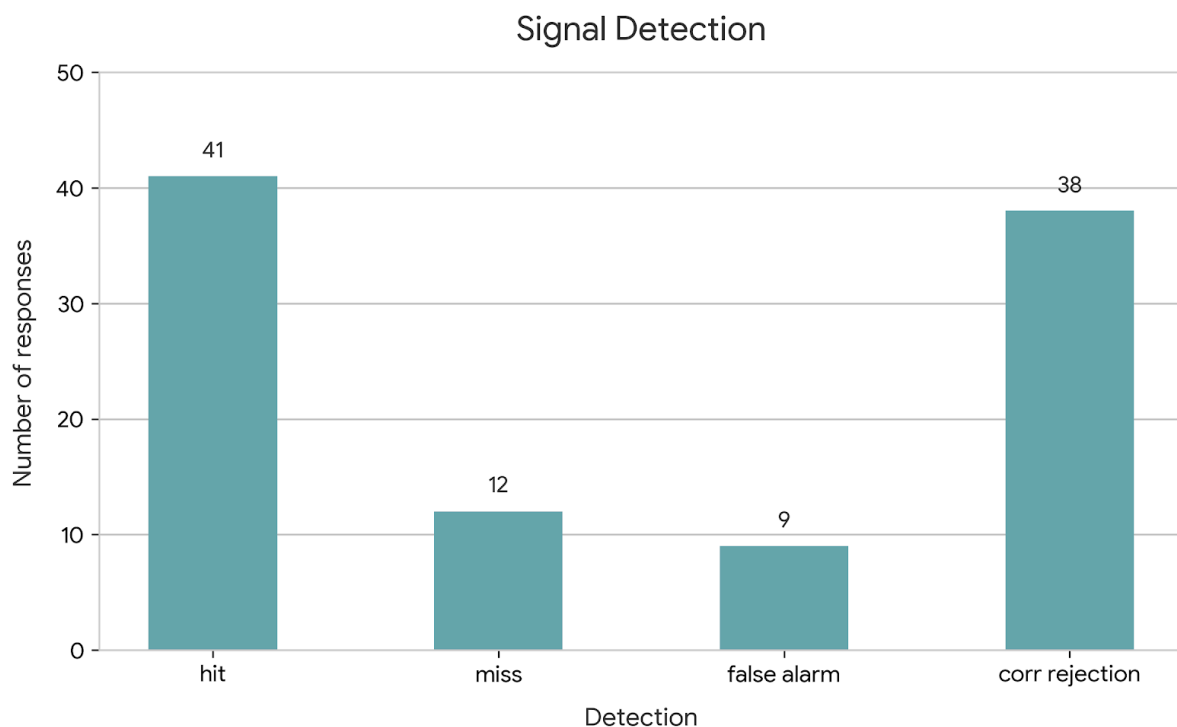
Each subject gave their informed consent before the experiment started. The task rules and

response keys were explained to participants verbally since the experimental program did not show textual instructions on the screen. They were told to assess the orientation of the grating patch by closely examining the visual cues on the screen. In particular, they were instructed to push the "down" arrow key when the grating was inclined lower or counterclockwise and the "up" arrow key when it was tilted upward or clockwise.

The experimental session began when participants attested to their comprehension of the task. Participants responded to a visual grating stimulus by hitting the corresponding arrow key on each trial. The software automatically recorded the participant's important decisions and accuracy as the experiment proceeded in a sequential manner across trials and stored it in the form of an excel spreadsheet.

Results

Signal Detection Theory (SDT) metrics were computed to assess orientation discrimination performance over hundred trials (fifty-three signal-present trials and forty-seven signal-absent trials). The participant successfully identified the target stimulus (tilt = 0) on all five trials (Hits = 41, Misses = 12), resulting in an unadjusted Hit rate of 0.77. However, the wrong response "up" was given on four trials (False Alarms = 9) and the right response "down" on one trial (Correct Rejection = 38) when the target was absent (tilt $\neq$ 0), yielding a False Alarm rate of 0.19 (19%).



Signal Condition	Respond "Up" (Target)	Respond "Down" (Tilt)	Total Trials
Signal Present (tilt = 0)	41 (Hit)	12 (Miss)	53
Signal Absent (tilt= 0)	9 (False Alarm)	38 (Correct Rejection)	47

The results of the signal detection analysis showed a response criteria (c) of -1.19 and a perceptual sensitivity score (d') of 1.62. The perceptual sensitivity (d') score thus indicates a moderate to strong ability to differentiate the vertical targets from the tilted non-targets, showing that the participant could reliably distinguish between line orientation across trials. On the other hand, the response criterion (-1.19) points to a liberal response bias, showing that the participant had a tendency to press the 'up' key when uncertain about the orientation of the grating.

Discussion

This experiment used Signal Detection Theory (SDT) measures to assess visual orientation discrimination performance across 100 trials. The individual obtained a response criterion of $c = -1.19$ and a perceptual sensitivity score of $d' = 1.62$. When taken as a whole, these findings show strong visual discrimination skills combined with a notably liberal response strategy.

The participant's visual system successfully identified the orientation differences between tilted non-target gratings and vertical target gratings, as indicated by the score of 1.62. A d' value greater than 1.0 is a recognized benchmark in Signal Detection Theory that indicates a participant has true perceptual skill rather than depending on guesswork.

Orientation-selective neurons in the primary visual cortex (V1) that fire preferentially to particular visual angles are physiologically responsible for visual orientation discrimination (Vázquez et al., 2000). The participant's capacity to discern these minute variations is consistent with previous research showing human accuracy in tasks involving cardinal orientation (Mikellidou et al., 2023).

The response criteria (c) represents the decision threshold utilized while making a choice under uncertainty, whereas d' indicates visual capacity. A significant liberal response bias is indicated by the very negative score of $c = -1.19$. The participant had a high Hit rate (77.36%) and a moderate False Alarm rate (19.15%) because they tended to push the "up" key

whenever unsure. This shows that the participant preferred reporting a target over determining if a tilt was present when visual evidence was unclear.

Compared to smaller pilot samples, expanding the experiment to 100 trials produced a considerably more accurate estimate of performance, removing the need for artificial proportion changes (e.g., adjusting 100% results). However, the degree of tilt presented across trials naturally affected performance in this task (e.g., greater 4° or 5° tilts were significantly simpler to notice than subtle 1° tilts).

Future versions of this experiment could build on these results by using trial-by-trial feedback to assist participants in maintaining a neutral judgment and fixed psychometric staircase approaches to identify the precise threshold angle where sensitivity diminishes.

References

Vázquez, P., Cano, M., & Acuña, C. (2000). Discrimination of line orientation in humans and monkeys. *Journal of neurophysiology*, 83(5), 2639–2648.
<https://doi.org/10.1152/jn.2000.83.5.2639>

Mikellidou, K., Lambrou, N., Georgiou, E., & Avraamides, M. (2023). Visual orientation discrimination skills are tightly linked with specific aspects of human intelligence. *PloS one*, 18(10), e0289590. <https://doi.org/10.1371/journal.pone.0289590>